



Outdoor air pollution, exhaled 8-isoprostane and current asthma in adults: the EGEA study

Anaïs Havet, Farid Zerimech, Margaux Sanchez, Valérie Siroux, Nicole Le Moual, Bert Brunekreef, Morgane Stempfelet, Nino Künzli, Bénédicte Jacquemin, Régis Matran, et al.

► To cite this version:

Anaïs Havet, Farid Zerimech, Margaux Sanchez, Valérie Siroux, Nicole Le Moual, et al.. Outdoor air pollution, exhaled 8-isoprostane and current asthma in adults: the EGEA study. *European Respiratory Journal*, 2018, 51 (4), Print 2018 Apr. <10.1183/13993003.02036-2017>. <inserm-01799461>

HAL Id: inserm-01799461

<https://inserm.hal.science/inserm-01799461v1>

Submitted on 24 May 2018

HAL is a multi-disciplinary open access archive for the deposit and dissemination of scientific research documents, whether they are published or not. The documents may come from teaching and research institutions in France or abroad, or from public or private research centers.

L'archive ouverte pluridisciplinaire **HAL**, est destinée au dépôt et à la diffusion de documents scientifiques de niveau recherche, publiés ou non, émanant des établissements d'enseignement et de recherche français ou étrangers, des laboratoires publics ou privés.



HAL Authorization

Outdoor air pollution, exhaled 8-isoprostane and current asthma in adults: the EGEA study

Anaïs Havet, Farid Zerimech, Margaux Sanchez, Valérie Siroux, Nicole Le Moual, Bert Brunekreef, Morgane Stempfelet, Nino Künzli, Bénédicte Jacquemin, Rachel Nadif, et al.

► To cite this version:

Anaïs Havet, Farid Zerimech, Margaux Sanchez, Valérie Siroux, Nicole Le Moual, et al.. Outdoor air pollution, exhaled 8-isoprostane and current asthma in adults: the EGEA study. European Respiratory Journal, European Respiratory Society, 2018, 51 (4), <10.1183/13993003.02036-2017>. <inserm-01799461>

HAL Id: inserm-01799461

<http://www.hal.inserm.fr/inserm-01799461>

Submitted on 24 May 2018

HAL is a multi-disciplinary open access archive for the deposit and dissemination of scientific research documents, whether they are published or not. The documents may come from teaching and research institutions in France or abroad, or from public or private research centers.

L'archive ouverte pluridisciplinaire **HAL**, est destinée au dépôt et à la diffusion de documents scientifiques de niveau recherche, publiés ou non, émanant des établissements d'enseignement et de recherche français ou étrangers, des laboratoires publics ou privés.

Outdoor air pollution, exhaled 8-isoprostanes and current asthma in adults: the EGEA study

Anaïs Havet^{1,2}, Farid Zerimech³, Margaux Sanchez⁴, Valérie Siroux⁵, Nicole Le Moual^{1,2}, Bert Brunekreef^{6,7}, Morgane Stempfelet⁸, Nino Künzli^{9,10}, Bénédicte Jacquemin^{1,2}, Régis Matran^{11*}, Rachel Nadif^{1,2*}

* These authors are joint last authors

1. INSERM, U1168, VIMA: Aging and chronic diseases. Epidemiological and public health approaches, Villejuif, France

2. Univ Versailles St-Quentin-en-Yvelines, UMRS 1168, Montigny-le-Bretonneux, France

3. CHU de Lille, Laboratoire de Biochimie et Biologie Moléculaire, Pôle de Biologie Pathologie Génétique, Lille, France

4. ISGlobal, Centre for Research in Environmental Epidemiology, Universitat Pompeu Fabra, CIBER Epidemiología y Salud Pública, Barcelona, Spain

5. Institute for Advanced Biosciences, Centre de recherche UGA-Inserm U1209 CNRS UMR 5309, équipe d'épidémiologie environnementale, Site Santé, Allée des Alpes, Grenoble, France

6. Institute for Risk Assessment Sciences, University Utrecht, Utrecht, the Netherlands

7. Julius Center for Health Sciences and Primary Care, University Medical Center Utrecht, Utrecht, the Netherlands;

8. Santé Publique France, Saint-Maurice, France

9. Swiss Tropical and Public Health Institute, Basel, Switzerland

10. University of Basel, Basel, Switzerland

11. Univ Lille et CHU de Lille, France

Address correspondence to A.Havet, INSERM UMR-S 1168 VIMA: Aging and chronic diseases. Epidemiological and public health approaches. 16 avenue Paul Vaillant Couturier, F-94807 VILLEJUIF Cedex. Telephone: +33145595073. E-mail: anais.havet@inserm.fr

Running title: outdoor air pollution, exhaled 8-isoprostanes and asthma.

Take home message: Exhaled 8-isoprostanes concentration is associated with both outdoor air pollution and current asthma in adults.

Plain language summary: Associations between outdoor air pollution and asthma in adults are still scarce, and the underlying biological mechanisms are poorly understood. Among adults, we studied the associations between 1) long-term exposure to outdoor air pollution and current asthma 2) exhaled 8-isoprostanes, a biomarker related to oxidative stress and current asthma, and 3) outdoor air pollution and exhaled 8-isoprostanes. We found for the first time associations between long-term exposures to outdoor air pollution estimated at individual level, exhaled 8-iso concentration and current asthma. Traffic intensity and O₃ exposure increased significantly the risk of current asthma. Exhaled 8-iso concentration was positively and significantly associated with current asthma. Among participants without asthma, exhaled 8-iso concentration increased significantly with PM_{2.5} exposure and decreased with O₃ and O₃-summer exposures. Results add new insights of a potential role of oxidative stress in the associations between outdoor air pollution and asthma in adults.

ABSTRACT

Associations between outdoor air pollution and asthma in adults are still scarce, and the underlying biological mechanisms are poorly understood.

To study the associations between 1) long-term exposure to outdoor air pollution and current asthma 2) exhaled 8-isoprostanes, a biomarker related to oxidative stress and current asthma, and 3) outdoor air pollution and exhaled 8-isoprostanes.

Cross-sectional analyses were conducted in 608 adults (39% with current asthma) from the first follow-up of the French case-control and family study on asthma (EGEA). The NO₂, NO_x, PM₁₀, PM_{2.5}, road traffic, and O₃ were assessed by ESCAPE FP7 and IFEN. Models took account city and familial dependence.

The risk of current asthma increased with traffic intensity (adjusted (a)OR=1.09 95%CI: [1.00, 1.18] per 5,000 vehicles/day), with O₃ exposure (aOR=2.04 [1.27, 3.29] per 10 µg/m³), and with exhaled 8-iso concentration (aOR=1.50 [1.06, 2.12] per 1 pg/mL). Among participants without asthma, exhaled 8-iso concentration increased with PM_{2.5} exposure (adjusted (a)β=0.23 95 CI [0.005, 0.46] per 5 µg/m³) and decreased with O₃ and O₃-summer exposures (aβ=-0.20 [-0.39, -0.01], aβ=-0.52 [-0.77, -0.26] per 10 µg/m³, respectively).

Results add new insights of a potential role of oxidative stress in the associations between outdoor air pollution and asthma in adults.

Introduction

According to the World Health Organization, outdoor air pollution caused 3.7 million deaths in the world in 2012, and 9 % of the total mortality in France was recently attributed to particulate matter with a diameter ≤ 2.5 microns (PM_{2.5}) [1]. The most studied pollutants in relation with health effects are nitrogen dioxide (NO₂), ozone (O₃) and particulate matter (PM₁₀ and PM_{2.5}). Evidence of the impact of outdoor air pollution on respiratory health is increasing [2] [3], and the associations between exposure to outdoor air pollution and asthma have been largely studied in children [4]. However, studies on associations between long-term exposure to outdoor air pollution and asthma among adults are still scarce [5].

One biological mechanism proposed to partly explain the association between outdoor air pollution and asthma is oxidative stress [6]. Oxidative stress is an imbalance between the increase to reactive oxygen species (ROS) and antioxidant response [7]. Among the biological markers related to oxidative stress, 8-isoprostanes (8-iso) are known as stable and specific products of lipid peroxidation [8]. Interestingly, 8-iso can be measured non-invasively in exhaled breath condensate (EBC), a fluid close to the lungs [7]. An increase of 8-iso concentration along with a decline of antioxidant defense can induce tissue damages and can contribute to pathophysiological changes as those seen in asthma [9, 10]. 8-iso are known to provoke airway hyperresponsiveness, to increase the production of mucus and to promote contraction of smooth muscles [10] [11]. In a recent systematic review, 8-iso concentration was found higher in adults with severe asthma than in those with mild-to-moderate asthma [12]. To date, epidemiological studies on the associations between long-term exposure to outdoor air pollution and asthma including the measurement of 8-iso in adults are lacking.

In the present paper, among adults from the Epidemiological study of the Genetic and Environmental factors of Asthma (EGEA), we first studied the associations between long-term exposure to outdoor air pollution (NO₂, NO_x, PM₁₀, PM_{2.5}, traffic load, traffic intensity,

O₃ and O₃-summer) and current asthma. Second, we studied the associations between 8-iso measured in EBC with both long-term exposure to outdoor air pollution and current asthma.

Methods

Study population

EGEA is a cohort study based on an initial group of asthma cases recruited in chest clinics from 5 French cities (1991-1995) along with their first-degree relatives, and a group of controls (<https://egeanet.vjf.inserm.fr/>). Child controls were recruited in surgery hospitals, and adult controls in electoral rolls and from surgery hospitals or check-up centers [13]. The protocol and descriptive characteristics have been described previously [13, 14], and inclusion criteria used to define asthmatic cases and controls were described in the online supplementary material. EGEA collection is certified ISO 9001 [15]. Ethical approval was obtained from the relevant institutional review board committees (Cochin Port-Royal Hospital and Necker-Enfants Malades Hospital, Paris). All participants signed a written informed consent.

The present analyses included 608 adults (≥ 16 years old) in the framework of the first follow-up (EGEA2, 2003-2007), with available data on current asthma, outdoor air pollution assessed by the European Study of Cohorts for Air Pollution Effects (ESCAPE) and exhaled 8-iso concentration (figure 1). Among the 608 participants, 5 had no pollution data from the French Institute for the Environment (IFEN) assessment. No significant differences were found between the participants included and those not included (n=963, see online supplementary table E1).

Respiratory phenotypes

At EGEA2, the participants with ever asthma answered positively to at least one of the two following questions “*Have you ever had attacks of breathlessness at rest with wheezing?*” or “*Have you ever had asthma attacks?*”, or were recruited as asthmatic cases at EGEA1.

Among participants with ever asthma, “*current asthma*” was defined by a report of respiratory symptoms (wheeze, nocturnal chest tightness, attacks of breathlessness following strenuous activity, at rest or at night time), or asthma attacks or use of inhaled and/or oral medicines because of breathing problems in the past twelve months [16] (see the online supplementary material for more details). Only participants with current asthma were included in the analyses because the phenotype “current asthma” which reflects the recent activity of the disease was more relevant than the phenotype “ever asthma” to study the associations between long-term exposure to outdoor air pollution, biological markers and asthma.

Exposure assessment

Thanks to ESCAPE and IFEN assessments, outdoor air pollution exposures (NO₂, NO_x, PM, O₃ and road traffic) were assigned to each participant’s residential address.

Annual air pollution levels of NO₂ and particulate matter (PM) were derived from ESCAPE standardized models (www.escapeproject.eu/manuals/). Briefly, the ESCAPE monitoring campaigns took place between 2009 and 2010, including 40 measurement sites for NO₂ and NO_x in Paris, Lyon, Grenoble and Marseille, and 20 PM measurement sites in Paris and Grenoble. Land-use regression (LUR) models were developed and two indicators of road traffic were also calculated. Traffic intensity on nearest road was defined as the number of motor vehicles circulating per day on nearest road to participant’s home, and was expressed in vehicles per day. Total traffic load was defined as the traffic load in all major roads based around a buffer of 100 meters from the participant’s home, and was expressed by traffic intensity multiplied by road length. The back-extrapolation is used to transfer the recent or

current LUR models to earlier years. In our study, the estimation of outdoor air pollution by ESCAPE took place after EGEA2, and accordingly we also analyzed the back-extrapolated pollution estimates in order to get a better temporality between outdoor air pollution with current asthma and EBC collection. Back-extrapolated pollution data were available for NO₂, NO_x in all cities, and for PM₁₀ in Paris.

In order to supplement the ESCAPE data set, we used O₃ and O₃-summer exposures from the IFEN (see the online supplementary material for more details). The O₃ estimate was the yearly mean of ozone level in 2004 for each participant at the residential address, and derived from a geo-statistical model as previously described [17]. The exposure of O₃-summer was assessed from the monthly means from April to September.

Measurement of exhaled 8-isoprostanes

Exhaled Breath Condensate (EBC) was collected at EGEA2 between 2003 and 2006 with an RTubeTM (Respiratory Research Inc., Charlottesville, VA, USA) according to a standardized method, as described previously [18]. Samples were immediately stored at -80°C. Exhaled 8-iso concentration was measured 5.3 to 9.4 years after EBC collection, with a specific enzyme immunoassay (EIA) kit (8-isoprostanes EIA kit Cayman Chemical, Ann Arbor, MI, USA) according to the manufacturer's protocol. Approximately 30% of exhaled 8-iso concentration was below the Limit of Detection (< LD, see the online supplementary material for more details).

Statistical analyses

The level of outdoor air pollution was described by city. Due to its skewed distribution, the exhaled 8-iso concentration was log-transformed.

Associations between outdoor air pollution and current asthma, and associations between

exhaled 8-iso concentration and current asthma were studied with logistic models. In order to study the association between outdoor air pollution and exhaled 8-iso concentration independently of asthma, linear regression models were first performed among participants without asthma and sensitivity analyses were conducted by: a) using back-extrapolated pollution estimates, b) studying the associations by city, and c) using bi-pollutant models (PM_{2.5} and O₃ or O_{3-summer}). Analyses with back-extrapolated pollution estimates were performed to get a better temporality between pollution and EBC collection. We also studied the associations between outdoor air pollution and exhaled 8-iso concentration in all participants, and among participants with current asthma.

All estimates were adjusted for age, sex, smoking habits. Further adjustments were conducted for body mass index (BMI), socio-professional category (SPC), and use of cleaning products which is an indicator of exposure to indoor pollution. In order to study only the road traffic effect, estimates of associations between road traffic and current asthma or exhaled 8-iso concentration were also adjusted for background NO₂. As the aim was to study the effect of long-term exposure to outdoor air pollution, we performed sensitivity analyses by excluding participants who lived less than one year at the same address (N=46).

In order to take into account the city-effect on outdoor air pollution levels, we performed meta-analyses including city-specific analyses from the Harvard School of Public Health's macro program (<https://www.hsph.harvard.edu/donna-spiegelman/software/metaanal>), and pooled analyses with random effect on city.

The above models have taken into account random effects on familial dependence. All the results are expressed for an increase of 1 pg/mL of exhaled 8-iso concentration, an increase of 20 µg/m³ of NO_x, 10 µg/m³ of NO₂, PM₁₀, O₃ and O_{3-summer} and 5 µg/m³ of PM_{2.5}. For traffic measures, the results of traffic load are expressed for four million vehicles × m/day in major roads within a 100-m buffer, and the results of traffic intensity are expressed for an increase

of 5,000 vehicles/day on major road. Statistical analyses were performed using SAS software, version 9.4 (SAS Institute, Inc., Cary, North Carolina, USA).

Results

The mean age of the 608 adults was 43 years, 47% were males, 39% had current asthma, 21% were current smokers, and 37 % were overweight ($\text{BMI} \geq 25 \text{ kg/m}^2$) (table 1). Participants with current asthma were younger, more often men, current smokers and unemployed, and had more often a secondary education level than those without asthma (table 1). Participants with current asthma had also a lower FEV_1 , higher airway hyperresponsiveness, higher allergic sensitization, higher IgE level, and higher exhaled 8-iso concentration than those without asthma.

The characteristics of participants were heterogeneous across cities (see the online supplementary table E2 and the online supplementary material for more details).

The pollutant and traffic levels were the highest in Paris, except for the PM_{10} level (online supplementary figures E1 and E2). The PM_{10} and $\text{PM}_{2.5}$ levels were above the values recommended by the WHO. The O_3 and $\text{O}_{3\text{-summer}}$ levels were the highest in Marseille (online supplementary figure E3). Positive and significant correlations were found between NO_2 , NO_x , PM_{10} , and $\text{PM}_{2.5}$ (correlation coefficient (r) $0.47 \leq r \leq 0.95$, all $P < 0.002$) whereas O_3 and $\text{O}_{3\text{-summer}}$ levels were negatively correlated with all pollutants ($-0.50 \leq r \leq -0.15$, all $P < 0.002$) (data not shown). The NO_2 level was significantly higher in managers and technicians, and $\text{O}_{3\text{-summer}}$ level was significantly higher in manual workers (data not shown).

The exhaled 8-iso concentration (25th;75th percentiles) was 3.16 (1.40;7.69) pg/mL in all participants and 3.97 (1.85;9.10) pg/mL among those with current asthma (table 1), and was 7 times higher in Paris than in other cities, was higher in women and decreased with age (online supplementary tables E2 and E3). No significant association was found between exhaled 8-iso

concentration and smoking expressed as current smoking habits, quantity of tobacco, or number of pack-years (all $P>0.80$, data not shown). No inter-plate variability was observed (data not shown) and no association was found between storage time and exhaled 8-iso concentration (regression coefficient=0.02, $P=0.39$).

Associations between outdoor air pollution and current asthma

Associations between outdoor air pollution and current asthma were not significantly heterogeneous between cities (Q test, $p\text{-value}>0.08$). In pooled analyses, the risk of current asthma increased significantly with traffic intensity (adjusted (a)OR=1.09 95%CI [1.00, 1.18]) and with O_3 exposure (aOR=2.04 [1.27, 3.29], table 2) whatever the adjustment. The results were similar after excluding participants who lived less than one year at the same address. Back-extrapolated exposure estimates gave similar results (online supplementary table E4).

Associations between exhaled 8-iso concentration and current asthma

A positive and significant association was found between exhaled 8-iso concentration expressed as a continuous variable and current asthma (aOR=1.50 [1.06, 2.12], figure 2). No significant association was found when exhaled 8-iso concentration was expressed as $<LD$ or $>LD$. Overall, the risk of current asthma increased significantly with exhaled 8-iso concentration expressed as $<LD$, $>LD$ and $\leq\text{median}$, $>LD$ and $>\text{median}$ (trend $p\text{-value}=0.05$). Exhaled 8-iso concentration was unrelated to duration of asthma (years), age of asthma onset expressed continuously (years) or in classes (all $P>0.40$). Further, no significant association was found between exhaled 8-iso concentration with FEV_1 % predicted continuously or in classes (all $P>0.60$) or allergic sensitization ($P=0.12$), (data not shown).

Associations between outdoor air pollution and exhaled 8-iso concentration among participants without asthma

Associations between outdoor air pollution and exhaled 8-iso concentration were not heterogeneous between cities (Q test, p -values >0.20). In pooled analyses, exhaled 8-iso concentration increased significantly with $PM_{2.5}$ exposure (adjusted $\alpha\beta=0.23$ [0.005, 0.46], and decreased with O_3 and $O_{3\text{-summer}}$ exposures ($\alpha\beta=-0.20$ [-0.39,-0.01] and $\alpha\beta=-0.52$ [-0.77,-0.26] whatever the adjustment, table 3). The results were similar after excluding participants who lived less than one year at the same address. Analyses performed with back-extrapolated data also gave similar results (online supplemental table E5). Furthermore, exhaled 8-iso concentration decreased significantly with O_3 and $O_{3\text{-summer}}$ exposures in Paris ($\alpha\beta=-0.22$ [-0.42,-0.02] and $\alpha\beta=-0.53$ [-0.90,-0.16], online supplementary tables E6). After excluding participants from Paris ($n=111$), no significant associations were observed between O_3 and $O_{3\text{-summer}}$ and exhaled 8-iso concentration (online supplementary table E7). In models adjusted for both $PM_{2.5}$ and O_3 , similar result was found only between $O_{3\text{-summer}}$ exposure and exhaled 8-iso concentration ($\alpha\beta=-0.59$ [-0.71,-0.47], online supplementary table E8). In models adjusted for both NO_2 and O_3 , the negative associations between O_3 and $O_{3\text{-summer}}$ and exhaled 8-iso concentration remained statistically significant (data not shown). In all participants, only $O_{3\text{-summer}}$ exposure was negatively and significantly associated with exhaled 8-iso concentration ($\alpha\beta=-0.33$ [-0.55,-0.11], data not shown). No association was found between outdoor air pollution and exhaled 8-iso concentration in participants with current asthma (data not shown).

Discussion

For the first time in adults, we found associations between long-term exposures to outdoor air pollution estimated at individual level, exhaled 8-iso concentration and current asthma.

267 Traffic intensity and O₃ exposure increased significantly the risk of current asthma. Exhaled
268 8-iso concentration was positively and significantly associated with current asthma. Among
269 participants without asthma, exhaled 8-iso concentration increased significantly with PM_{2.5}
270 exposure and decreased with O₃ and O_{3-summer} exposures.

271 Participants with asthma included in the present analyses were mostly recruited in chest
272 clinics as asthma cases, with a careful procedure set up to include true asthmatics using
273 standardized and validated questionnaires. Others were recruited as first-degree relatives of
274 asthmatic cases based on answers to questions on asthma diagnosis. This leads to a group of
275 asthmatics with a wide range of disease expression. In our cross-sectional analyses, cause and
276 consequence cannot be disentangled. It was not possible to study the associations between
277 outdoor air pollution and exhaled 8-iso concentration with the incidence of asthma because
278 only 30 new cases of asthma were reported at EGEA2. But, there is clearly a need for further
279 research to confirm the associations and to clarify its causal underpinnings. Regarding
280 exposure assessment, LUR models are well-adapted to take into account the spatial variation
281 of NO₂, PM_{2.5} [19] and NO_x [20], and ESCAPE resolution is accurate to estimate the
282 exposure to markers of road traffic which have a spatial heterogeneity. In addition, IFEN
283 resolution is larger than ESCAPE but suitable for O₃ and O_{3-summer} which are homogeneous
284 over long distances [21]. We acknowledge that a weakness of our study is the non-compliance
285 of the temporality because outdoor air pollution was estimated by ESCAPE between 2009 and
286 2010 whereas the collection of EBC and phenotype “current asthma” took place at EGEA2
287 between 2003 and 2007. To get a better temporality in our analyses, we used the back-
288 extrapolated pollution estimates which were back-extrapolated at participants’ residential
289 address at EGEA2, and found similar results. Both back-extrapolated and non-back-
290 extrapolated estimates were highly and significantly correlated in our study (correlation
291 coefficients ≥ 0.98 , $P < 0.001$) as previously reported by Beelen *et al.* in the same ESCAPE

project and for a longer period [22]. Previously in ESCAPE study, associations between NO₂ back-extrapolated estimates and asthma incidence were similar to those with non-back-extrapolated estimates [23]. We aimed to study the impact of long-term exposure to outdoor air pollution, and therefore conducted sensitivity analyses by excluding participants who lived less one year at the same residential address that did not change our conclusions. We cannot exclude that some non-differential misclassification of pollution exposure may have occurred because the time-activity patterns of participants were not available in our analyses, but in this case, it would lead to bias towards the null. Furthermore, IFEN resolution can better represent daily participant's exposures to O₃ and O₃-summer, at least for those whom work was close to home. We could not take into account all indoor environmental factors; however, we found similar results after further adjustment for domestic exposure to cleaning products. Furthermore, the adjustment for the socioeconomic position potentially associated with pollutant exposures [24] and asthma gave also similar results. The EGEA study is a case-control and family study. Participants from the same family share genetic background, and also socio-economic and lifestyle factors, that could be associated with the exposure to outdoor air pollution or the asthma risk. We therefore took into account familial dependence through random effects in mixed models. Sub-groups analyses suffer from a lack of power but, as best as possible, we used the most suitable statistical models. Finally, the EIA method has been preferred for dosing exhaled 8-iso concentration rather the GCMS method because the former is better adapted to analyze a larger number of samples as in our study.

We found that traffic intensity and O₃ exposure increased the risk of current asthma. Our results add evidence of the impact of long-term exposure to outdoor air pollution on asthma in adults. Our results are partly in accordance with those of a recent study showing that traffic exposure but not NO₂ exposure, assessed by satellite-based LUR model at residential addresses, was positively associated with current asthma in 1367 adults [25]. We also found a

positive association between O₃ exposure and current asthma. To our knowledge, the literature assessing outdoor air pollution effect on asthma focused mainly on other asthma phenotypes such as asthma onset, asthma severity, or asthma control. A study conducted in California has reported that long-term exposure to O₃ was associated with development of asthma in adult males [26]. Previously in the EGEA study, long-term exposure to O₃ estimated by IFEN was associated with asthma severity [17] and with uncontrolled asthma [27]. Recently, a cohort showed that asthmatic adults exposed to O₃ had a greater risk to develop asthma-chronic obstructive pulmonary disease [28]. Interestingly, various asthma phenotypes were studied such as asthma-onset which reflects the initiation of the disease, and severity and control of asthma which are linked to manifestations of the disease. Asthma reflects both ever asthma and current asthma, and the participants with ever asthma had not necessarily a current asthma. In the EGEA study, the phenotype “current asthma” was defined by the report of respiratory symptoms or asthma attacks or use of inhaled and/or oral medicines because of breathing problems in the past twelve months. To study the associations between long-term exposure to outdoor air pollution, biological markers and asthma, the phenotype “current asthma” was more relevant than the phenotype “ever asthma” because it reflects the recent activity of the disease. Overall, all these findings add evidence of associations between outdoor air pollution and asthma in adults, whatever the studied phenotypes.

This study adds new insights into a potential role of oxidative stress in the associations between long-term exposure to outdoor air pollution and asthma in adults. We reported a significant association between exhaled 8-iso concentration and current asthma after adjustment for age, sex, smoking habits and body mass index. We investigated whether other asthma characteristics could explain this association, but we did not found any significant association between exhaled 8-iso concentration with duration of asthma, age of asthma onset,

lung function, or allergic sensitization. Our results added new evidence to the previous associations reported in the literature with asthma severity and asthma control [12]. We found for the first time that exhaled 8-iso concentration was positively associated with PM_{2.5} exposure in participants without asthma. Unfortunately, we did not have back-extrapolated data for PM_{2.5}. In France, PM_{2.5} level has overall decreased between 2002 and 2012 and the association between PM_{2.5} exposure and exhaled 8-iso concentration may be underestimated [29]. The literature mainly focused on short-term exposure to outdoor air pollution and on other biological compartments. In fact, short-term exposure to PM_{2.5} was previously found to be positively associated with EBC 8-iso concentration among healthy adolescents [30], and with 8-iso concentration in urine among adults [31]. We also found counterintuitive negative associations between O₃ and O_{3-summer} exposures and exhaled 8-iso concentration. Our results are not in accordance with those of a previous study showing that O₃ exposure assigned to residential location was associated with higher plasma 8-iso concentration in 120 healthy students [32]. The inconsistency with our findings may be partially attributed to differences in the study design, in the biological compartment, and in the spatial resolution which was less accurate than in our study. We showed that O₃ exposure and exhaled 8-iso concentration increased the risk of current asthma, and we did not expect negative associations between O₃ and O_{3-summer} exposures and exhaled 8-iso concentration. We investigated more thoroughly why these associations were negative. The analyses conducted by city showed that Paris heavily weighted the negative associations between O₃ and O_{3-summer} exposures and exhaled 8-iso concentration. Previously, a study has also reported a negative association between short-term exposure to O₃ and exhaled 8-iso in New Yorker's adolescents [33]. Another explanation is that O₃ is a secondary pollutant mainly produces through complex chemical reactions from NO₂. In our study, O₃ and NO₂ levels were highly and negatively correlated, and NO₂ exposure was also positively associated with exhaled 8-iso concentration but not significantly.

One hypothesis is likely that the positive association between NO₂ and exhaled 8-iso partly explained the negative associations observed for O₃ and O₃-summer. Overall, the specific effects of pollutants are hard to disentangle even in bi-pollutant models given the strong correlations between pollutants. It is interesting to note that participants from Paris had particular characteristics as an exhaled 8-iso concentration 7 times higher, were exposed to higher levels of road traffic, NO₂ and PM, and to lower levels of O₃ and O₃-summer than in other cities, suggesting that Parisians may have particular characteristics that could partly explain the negative associations between ozone and exhaled 8-iso concentration. O₃ and O₃-summer were negatively correlated with PM_{2.5}, but the negative association between O₃-summer exposure and exhaled 8-iso concentration remained significant in a bi-pollutant model. Previously in EGEA adults, O₃ assessed by IFEN was also found to be negatively associated with the EBC total nitrites/nitrates level, a biological marker of nitrosative stress [34]. In the literature, complex interplays between nitrosative and oxidative stress pathways have been reported [35], including a reciprocal regulation. An alternative explanation is that our results were likely due to other factors that are not included in the present analyses. In all participants, only a negative and significant association between O₃-summer exposure and exhaled 8-iso concentration was found, and no association was found in participants with current asthma. In a directed acyclic graph (DAG), the covariate “current asthma” may be a collider, which makes the interpretation of our results even more difficult in a context of cross-sectional analyses [36].

Several biological mechanisms by which outdoor air pollution may be associated with asthma have been suggested in previous reviews [37, 38]. Outdoor air pollution exposure may increase oxidative stress in airways through the production of reactive oxygen species and local inflammation [9]. The PM can be supports for allergens and their small size gives them a large surface related to oxidative potential [39]. To disentangle the complexity of the

associations between asthma and oxidative stress, and 8-iso in particular, there is a need for longitudinal epidemiological studies. Overall, our results add new insights of a potential role of oxidative stress in the association between long-term exposure to outdoor air pollution and asthma in adults. To better understand the underlying biological pathways between outdoor air pollution and asthma, future epidemiological studies should use individual portable sensors in order to improve the pollution assessment, study the different windows of exposure and try to collect multiple exposures in order to identify exposure profiles through clustering methods.

Conclusions

We found that traffic intensity, O₃ exposure and exhaled 8-iso concentration increased the risk of current asthma, and that among participants without asthma exhaled 8-iso concentration increased with PM_{2.5} exposure, and decreased with O₃ and O₃-summer exposures. Exhaled 8-iso seems to be an interesting oxidative stress-related biomarker adapted to epidemiological studies. Overall, our study adds new insights in the associations between long-term exposure to outdoor air pollution and asthma in adults, and suggests that oxidative stress may partly explained such associations. Longitudinal studies with larger samples are now needed to confirm such results.

AKNOWLEDGMENTS

EGEA cooperative group. Coordination: V Siroux (epidemiology, PI since 2013); F Demenais (genetics); I Pin (clinical aspects); R Nadif (biology); F Kauffmann (PI 1992-2012). **Respiratory epidemiology:** Inserm ex-U 700, Paris: M Korobaef (Egea1), F Neukirch (Egea1); Inserm ex-U 707, Paris: I Annesi-Maesano (Egea1-2); Inserm ex-U 1018, Villejuif: F Kauffmann, MP Oryszczyn (Egea1-2); Inserm U 1168, Villejuif: N Le Moual, R Nadif, R Varraso; Inserm U 1209 Grenoble: V Siroux. **Genetics:** Inserm ex-U 393, Paris: J Feingold; Inserm U 946, Paris: E Bouzigon, F Demenais, MH Dizier; CNG, Evry: I Gut (now CNAG, Barcelona, Spain), M Lathrop (now Univ McGill, Montreal, Canada). **Clinical centers:** Grenoble: I Pin, C Pison; Lyon: D Ecochard (Egea1), F Gormand, Y Pacheco; Marseille: D Charpin (Egea1), D Vervloet (Egea1-2); Montpellier: J Bousquet; Paris Cochin: A Lockhart (Egea1), R Matran (now in Lille); Paris Necker: E Paty (Egea1-2), P Scheinmann (Egea1-2); Paris-Trousseau: A Grimfeld (Egea1-2), J Just. **Data and quality management:** Inserm ex-U155 (Egea1): J Hochez; Inserm U 1168, Villejuif: N Le Moual; Inserm ex-U780: C Ravault (Egea1-2); Inserm ex-U794: N Chateigner (Egea1-2); Grenoble: J Quentin (Egea1-2).

The authors thank all those who participated to the setting of the study and on the various aspects of the examinations involved: interviewers, technicians for lung function testing and skin prick tests, blood sampling, IgE determinations, coders, those involved in quality control, data and sample management and all those who supervised the study in all centers. The authors are grateful to the three CIC-Inserm of Necker, Grenoble and Marseille who supported the study and in which participants were examined. They are also grateful to the biobanks in Lille (CIC-Inserm), and at Annemasse (Etablissement français du sang) where biological samples are stored. They are indebted to all the individuals who participated,

without whom the study would not have been possible. The authors thank N. Jeannée from
Géovariances and all those from the French Institute for Environment, and particularly M. Ba,
for their work on the geo-spatial models.

FUNDING

Research was funded in part by the National Hospital program of clinical research (PHRC-
national 2012, EvAdA), ANR-CES-2009, Region Nord Pas-de-Calais, Merck Sharp &
Dohme (MSD), the GA2LEN project, Global Allergy and Asthma European Network, and the
Fonds AGIR pour les maladies chroniques. ESCAPE (FP7/2007-2011, Grant Nr.211250).
Anaïs Havet is financed by the university hospital center of Lille.

COMPETING FINANCIAL INTEREST

Valérie Siroux reports personal fees (speaker honorarium) from TEVA, AstraZeneca and
Novartis, outside the submitted work.

467

468 **References**

- 469 1. Pascal M, De Crouy Chanel P, Wagner V, Corso M, Tillier C, Bentayeb M, Blanchard
470 M, Cochet A, Pascal L, Host S, Gorla S, Le Tertre A, Chatignoux E, Ung A, Beaudeau
471 P, Medina S. The mortality impacts of fine particles in France. *Sci. Total Environ* 2016;
472 571: 416-25 Available from: <http://dx.doi.org/10.1016/j.scitotenv.2016.06.213>.
- 473 2. Kelly FJ, Fussell JC. Air pollution and airway disease. *Clin. Exp. Allergy* 2011; 41:
474 1059–1071
- 475 3. Mannucci PM, Harari S, Martinelli I, Franchini M. Effects on health of air pollution: a
476 narrative review. *Intern. Emerg. Med.* Springer Milan; 2015; 10: 657–662.
- 477 4. Goldizen FC, Sly PD, Knibbs LD. Respiratory effects of air pollution on children.
478 *Pediatr. Pulmonol.* 2016; 51: 94–108.
- 479 5. Jacquemin B, Schikowski T, Carsin A, Hansell A, Krämer U, Sunyer J, Probst-Hensch
480 N, Kauffmann F, Künzli N. The role of air pollution in adult-onset asthma: A review of
481 the current evidence. *Semin. Respir. Crit. Care Med* 2012; 33: 606–619 Available from:
482 <http://dx.doi.org/10.1055/s-0032-1325191>.
- 483 6. Guarnieri M, Balmes JR. Outdoor air pollution and asthma. *Lancet* 2015; 383: 1581–
484 1592.
- 485 7. Czerska M, Zieliński M, Gromadzińska J. Isoprostanes – A novel major group of
486 oxidative stress markers. *Int. J. Occup. Med. Environ. Health* 2016; 29: 179–190.
- 487 8. Roberts LJ, Morrow JD. Measurement of F2-isoprostanes as an index of oxidative
488 stress in vivo. *Free Radic. Biol. Med.* 2000; 28: 505–513.
- 489 9. Kelly FJ. Oxidative stress: its role in air pollution and adverse health effects. *Occup.*
490 *Environ. Med.* 2003; 60: 612–616.

- 491 10. Wood L, Gibson PG, Garg ML. Biomarkers of lipid peroxidation, airway inflammation
492 and asthma. *Eur. Respir. J.* 2003; 21: 177–186 Available from:
493 <http://erj.ersjournals.com/cgi/doi/10.1183/09031936.03.00017003>.
- 494 11. Janssen LJ. The pulmonary biology of isoprostanes. *Chem. physocs Lipids* 2004; 128:
495 101–116.
- 496 12. Aldakheel FM, Thomas PS, Bourke JE, Matheson MC, Dharmage SC, Lowe AJ.
497 Relationships between adult asthma and oxidative stress markers and pH in exhaled
498 breath condensate: a systematic review. *Allergy* 2016; 71: 741–757 Available from:
499 <http://www.ncbi.nlm.nih.gov/pubmed/26896172>.
- 500 13. Kauffmann F, Dizier MH. EGEA (Epidemiological study on the Genetics and
501 Environment of Asthma, bronchial hyperresponsiveness and atopy)--design issues. *Clin.*
502 *Exp. Allergy J. Br. Soc. Allergy Clin. Immunol.* 1995; 25 Suppl 2: 19–22.
- 503 14. Kauffmann F, Dizier M-H, Annesi-Maesano I, Bousquet J, Charpin D, Demenais F,
504 Ecochard D, Feingold J, Gormand F, Grimfeld A, Lathrop M, Matran R, Neukirch F,
505 Paty E, Pin I, Pison C, Scheinmann P, Vervloet D, Lockhart A. EGEA (Epidemiological
506 study on the Genetics and Environment of Asthma, bronchial hyperresponsiveness and
507 atopy) - descriptive characteristics. *Clin. Exp. Allergy* 1999; 29: 17–21.
- 508 15. Nadif R, Bouzigon E, Le Moual N, Siroux V. EGEA Collection: a biobank devoted to
509 asthma and asthma-related phenotypes. *Open J. Bioresour.* 2017; 4: 5.
- 510 16. Siroux V, Boudier A, Bousquet J, Bresson JL, Cracowski JL, Ferran J, Gormand F, Just
511 J, Le Moual N, Morange S, Nadif R, Oryszczyn MP, Pison C, Scheinmann P, Varraso
512 R, Vervloet D, Pin I, Kauffmann F. Phenotypic determinants of uncontrolled asthma. *J.*
513 *Allergy Clin. Immunol.* 2009; 124: 681-7.
- 514 17. Rage E, Siroux V, Künzli N, Pin I, Kauffmann F. Air pollution and asthma severity in
515 adults. *Occup. Environ. Med.* 2009; 66: 182–188 Available from:

<http://dx.doi.org/10.1136/oem.2007.038349>

18. Nadif R, Rava M, Decoster B, Huyvaert H, Le Moual N, Bousquet J, Siroux V, Varraso R, Pin I, Zerimech F, Matran R. Exhaled nitric oxide, nitrite/nitrate levels, allergy, rhinitis and asthma in the EGEA study. *Eur. Respir. J.* 2014; 44: 351–360.
19. Wang M, Beelen R, Bellander T, Birk M, Cesaroni G, Cirach M, Cyrus J, de Hoogh K, Declercq C, Dimakopoulou K, Eeftens M, Eriksen KT, Forastiere F, Galassi C, Grivas G, Heinrich J, Hoffmann B, Ineichen A, Korek M, Lanki T, Lindley S, Modig L, Mölter A, Nafstad P, Nieuwenhuijsen MJ, Nystad W, Olsson D, Raaschou-Nielsen O, Ragettli M, Ranzi A, et al. Performance of multi-city land use regression models for nitrogen dioxide and fine particles. *Environ. Health Perspect.* 2014; 122: 843–849.
20. Beelen R, Hoek G, Vienneau D, Eeftens M, Dimakopoulou K, Pedeli X, Tsai MY, Künzli N, Schikowski T, Marcon A, Eriksen KT, Raaschou-Nielsen O, Stephanou E, Patelarou E, Lanki T, Yli-Tuomi T, Declercq C, Falq G, Stempfelet M, Birk M, Cyrus J, von Klot S, Nádor G, Varró MJ, Dedele A, Gražulevičiene R, Mölter A, Lindley S, Madsen C, Cesaroni G, et al. Development of NO₂ and NO_x land use regression models for estimating air pollution exposure in 36 study areas in Europe - The ESCAPE project. *Atmos. Environ.* 2013; 72: 10–23.
21. Rava M, Varraso R, Decoster B, Huyvaert H, Le Moual N, Jacquemin B, Künzli N, Kauffmann F, Zerimech F, Matran R, Nadif R. Plasma and exhaled breath condensate nitrite-nitrate level in relation to environmental exposures in adults in the EGEA study. *Nitric Oxide* 2012; 27: 169–175 Available from: <http://dx.doi.org/10.1016/j.niox.2012.06.007>.
22. Beelen R, Hoek G, Fischer P, Brandt PA van den, Brunekreef B. Estimated long-term outdoor air pollution concentrations in a cohort study. *Atmos. Environ.* 2007; 41: 1343–1358.

23. Jacquemin B, Siroux V, Sanchez M, Carsin A, Schikowski T, Adam M, Bellisario V, Buschka A, Bono R, Brunekreef B, Yutong C, Cirach M, Clavel-Chapelon F, Declercq C, De Marco R, De Nazelle A, Ducret-Stich R E, Ferretti V V, Gerbase M W, Hardy R, Heinrich J, Janson C, Jarvis D, Kanaani Z, Keidel D, Kuch D, Le Moual N, Nieuwenhuijsen M S, Marcon A, Modig L et al. Ambient Air Pollution and Adult Asthma Incidence in Six European Cohorts (ESCAPE). *Environ. Heal. Perspect.* 2015; 123: 613–622.
24. Temam S, Burte E, Adam M, Antó JM, Basagaña X, Bousquet J, Carsin A-E, Galobardes B, Keidel D, Künzli N, Le Moual N, Sanchez M, Sunyer J, Bono R, Brunekreef B, Heinrich J, de Hoogh K, Jarvis D, Marcon A, Modig L, Nadif R, Nieuwenhuijsen M, Pin I, Siroux V, Stempfelet M, Tsai M-Y, Probst-Hensch N, Jacquemin B. Socioeconomic position and outdoor nitrogen dioxide (NO₂) exposure in Western Europe: A multi-city analysis. *Environ. Int* 2017; 101: 117–124 Available from: <http://linkinghub.elsevier.com/retrieve/pii/S0160412016303981>.
25. Bowatte G, Lodge CJ, Knibbs LD, Lowe AJ, Erbas B, Dennekamp M, Marks GB, Giles G, Morrison S, Thompson B, Thomas PS, Hui J, Perret JL, Abramson MJ, Walters H, Matheson MC, Dharmage SC. Traffic-related air pollution exposure is associated with allergic sensitization, asthma, and poor lung function in middle age. *J. Allergy Clin. Immunol* 2016; 139: 1–9 Available from: <http://dx.doi.org/10.1016/j.jaci.2016.05.008>.
26. McDonnell WF, Abbey DE, Nishino N, Lebowitz MD. Long-term ambient ozone concentration and the incidence of asthma in nonsmoking adults: The Ahsmog study. *Environ. Res.* 1999; 80: 110–121.
27. Jacquemin B, Kauffmann F, Pin I, Le Moual N, Bousquet J, Gormand F, Just J, Nadif R, Pison C, Vervloet D, Künzli N, Siroux V. Air pollution and asthma control in the

Epidemiological study on the Genetics and Environment of Asthma. *J. Epidemiol. Community Heal.* 2012; 66: 796–802.

28. To T, Zhu J, Larsen K, Simatovic J, Feldman L, Ryckman K, Gershon A, Diane Loughheed M, Licskai C, Chen H, Villeneuve PJ, Crighton E, Su Y, Sadatsafavi M, Williams D, Carlsten C. Progression from asthma to chronic obstructive pulmonary disease is air pollution a risk factor? *Am. J. Respir. Crit. Care Med.* 2016; 194: 429–438.

29. Airparif. Évolution qualité de l'air à Paris entre 2002-2012. Available from : https://www.airparif.asso.fr/_pdf/publications/rapport-pdp-130703.pdf

30. Hashemzadeh B, Idani E, Goudarzi G, Ankali KA, Sakhvidi MJZ, Akbar Babaei A, Hashemzadeh H, Vosoughi M, Mohammadi MJ, Neisi A. Effects of PM_{2.5} and NO₂ on the 8-isoprostane and lung function indices of FVC and FEV₁ in students of Ahvaz city, Iran. *Saudi J. Biol. Sci* 2016 Available from: <http://dx.doi.org/10.1016/j.sjbs.2016.11.008>.

31. Li W, Wilker EH, Dorans KS, Rice MB, Schwartz J, Coull BA, Koutrakis P, Gold DR, Keaney JF, Lin H, Vasan RS, Benjamin EJ, Mittleman MA. Short - Term Exposure to Air Pollution and Biomarkers of Oxidative Stress: The Framingham Heart Study. *J. Am. Heart Assoc* 2016; 5: e002742 Available from: <http://jaha.ahajournals.org/lookup/doi/10.1161/JAHA.115.002742>.

32. Chen C, Arjomandi M, Balmes J, Tager I, Holland N. Effects of chronic and acute ozone exposure on lipid peroxidation and antioxidant capacity in healthy young adults. *Environ. Health Perspect.* 2007; 115: 1732–1737.

33. Patel MM, Chillrud SN, Deepti KC, Ross JM, Kinney PL. Traffic-related air pollutants and exhaled markers of airway inflammation and oxidative stress in New York City adolescents. *Environ. Res.* 2013; 121: 71–78.

34. Rava M, Varraso R, Decoster B, Huyvaert H, Le Moual N, Jacquemin B, Künzli N, Kauffmann F, Zerimech F, Matran R, Nadif R. Plasma and exhaled breath condensate nitrite-nitrate level in relation to environmental exposures in adults in the EGEA study. *Nitric Oxide* 2012; 27: 169–175 Available from: <http://dx.doi.org/10.1016/j.niox.2012.06.007>.
35. Cuzzocrea S, Salvemini D. Molecular mechanisms involved in the reciprocal regulation of cyclooxygenase and nitric oxide synthase enzymes. *Kidney Int* 2007; 71: 290–297 Available from: <http://dx.doi.org/10.1038/sj.ki.5002058>.
36. Suttorp MM, Siegerink B, Jager KJ, Zoccali C, Dekker FW. Graphical presentation of confounding in directed acyclic graphs. 2015; 30: 1418–1423.
37. Gowers AM, Cullinan P, Ayres JG, Anderson HR, Strachan DP, Holgate ST, Mills IC, Maynard RL. Does outdoor air pollution induce new cases of asthma? Biological plausibility and evidence ; a review. *Respirology* 2012; 17: 887–898.
38. Bernstein JA, Alexis N, Barnes C, Bernstein IL, Bernstein JA, Nel A, Peden D, Diaz-Sanchez D, Tarlo SM, Williams PB. Health effects of air pollution. *Environ. Occup. Respir. Disord.* 2004; 1116–1123.
39. Beck-speier I, Dayal N, Karg E, Maier KL, Schumann G, Schulz H, Semmler M, Takenaka S, Stettmaier K, Bors W, Ghio A, Samet JM, Heyder J. Oxidative stress and lipid mediators induced in alveolar macrophages by ultrafine particles. *Free Radic. Biol. Med.* 2005; 38: 1080–1092.

Table 1. Characteristics of adult participants according to asthma status.

Characteristics	In all participants n=608	Participants without asthma n=368	Participants with current asthma n=240	p-value
Age (years), mean $\pm$ SD	42.5 $\pm$ 17.2	45.5 $\pm$ 26.3	37.9 $\pm$ 17.5	<0.0001⁺
Sex, Men, n (%)	288 (47.4)	160 (43.5)	128 (53.3)	0.02⁺
Smoking habits, n (%)	312 (51.3)	189 (51.4)	123 (51.3)	
Never smokers	168 (27.6)	113 (30.7)	55 (22.9)	0.02⁺
Ex-smokers	128 (21.1)	66 (17.9)	62 (25.8)	
Current smokers				
BMI (kg/m ²), n (%)	70 (11.5)	38 (10.3)	32 (13.3)	
<20	316 (52.0)	193 (52.4)	123 (51.3)	
[20-25[	165 (27.1)	104 (28.3)	61 (25.4)	0.62
[25-30[	57 (9.4)	33 (9.0)	24 (10.0)	
≥ 30				
Socio-professional category, n (%)	n=605	n=366	n=239	
Unemployed	72 (11.9)	29 (7.9)	43 (18.0)	
Manager	212 (35.0)	138 (37.7)	74 (31.0)	0.001⁺
Technician	248 (41.0)	149 (40.7)	99 (41.4)	
Manual worker	73 (12.1)	50 (13.7)	23 (9.6)	
Number of years at the same residential address, mean $\pm$ SD (min;max)	n=607 12.4 $\pm$ 10.7 (0;48)	n=366 13.0 $\pm$ 10.9 (0;48)	n=239 11.4 $\pm$ 10.4 (0;46)	0.07[†]
Asthma, n (%)				
Current	240 (39.5)	/	/	
Duration of asthma (years), mean $\pm$ SD (min;max)	/	/	n=229 15.9 $\pm$ 11.6 (0;59.6)	
Age of asthma onset (years)			n=228	
≤ 4	/	/	80 (35.1)	
[4-16]			79 (34.7)	
>16			69 (30.3)	
FEV1 % predicted, mean $\pm$ SD	103 $\pm$ 18.9	108 $\pm$ 17.3	95.2 $\pm$ 18.5	<0.0001⁺
Allergic sensitization [‡] , n (%)	n=602 327 (54.3)	(n=363) 137 (37.7)	n=239 190 (79.5)	<0.0001⁺
Exhaled 8-iso concentration, pg/mL, GM (q1;q3)	3.16 (1.41;7.69)	2.69 (1.14;6.79)	3.97 (1.85;9.10)	0.004⁺
Exhaled 8-iso concentration >LD, n (%)	431 (70.9)	253 (68.8)	178 (74.2)	0.15

BMI, body mass index; 8-iso, 8-isoprostanes; GM, geometric mean; q1 and q3, the 25th and the 75th percentiles of the GM; LD, limit of detection of the 8-iso concentration. FEV₁, forced expiratory volume; [†]: defined by at least one wheal ≥ 3 mm to 12 tested allergens; [‡] Results in bold represent significant results (p-values ≤ 0.05).

Table 2. Associations between outdoor air pollution and current asthma.

		NO ₂	NO _x	PM ₁₀	PM _{2.5}	Total traffic load on major roads in a 100-m buffer of the home [#]	Traffic intensity at the road nearest to a participant's home [#]	O ₃	O ₃ -summer
Model 1	n	608	608	437	437	605	605	603	603
	OR crude (95% CI)	0.99 (0.87,1.15)	1.05 (0.92,1.19)	1.10 (0.65,1.85)	0.89 (0.54,1.45)	1.16 (0.96,1.41)	1.09 (1.00,1.19)[†]	2.01 (1.26,3.23)[†]	1.29 (0.68,2.46)
Model 2	n	608	608	437	437	608	608	603	603
	OR adjusted (95% CI)	0.99 (0.86,1.14)	1.04 (0.91,1.18)	1.05 (0.62,1.79)	0.82 (0.49,1.36)	1.15 (0.95,1.38)	1.09 (1.00,1.19)[†]	1.93 (1.21,3.09)[†]	1.10 (0.67,1.81)
Model 3	n	605	605	436	436	605	605	600	600
	OR adjusted (95% CI)	1.00 (0.87,1.16)	1.05 (0.92,1.19)	1.05 (0.61,1.81)	0.84 (0.50,1.39)	1.15 (0.95,1.39)	1.10 (1.00,1.20)[†]	1.89 (1.19,3.02)[†]	1.10 (0.67,1.81)
Model 4	n	603	603	435	435	603	603	598	598
	OR adjusted (95% CI)	0.98 (0.85,1.14)	1.03 (0.90,1.17)	1.03 (0.59,1.80)	0.82 (0.49,1.39)	1.14 (0.94,1.37)	1.09 (1.00,1.18)[†]	2.04 (1.27,3.29)[†]	1.24 (0.71,2.18)
Model 5	n	557	557	400	400	557	557	553	553
	OR adjusted (95% CI)	0.99 (0.86,1.16)	1.04 (0.91,1.19)	1.03 (0.57,1.85)	0.89 (0.52,1.53)	1.18 (0.97,1.45)	1.10 (1.00,1.19)[†]	1.67 (1.06,2.63)[†]	1.06 (0.77,1.45)

Data are presented as OR and 95% confidence intervals (95% CI) with the participants without asthma as reference. The logistic models were conducted with random effects on familial dependence (level 2) and city (level 3).

Results are expressed per 20 µg/m³ increase of NO_x exposure, per 10 µg/m³ increase of NO₂, PM₁₀, O₃ and O₃-summer exposures, per 5 µg/m³ increase of PM_{2.5} exposure, per 4.000.000 vehicles x m /day increase of total traffic load and per 5.000 vehicles/day increase of traffic intensity.

Model 1: unadjusted; Model 2: adjusted for age, sex, smoking habits and body mass index. Model 3: adjusted for age, sex, smoking habits, body mass index and socio-professional category. Model 4: adjusted for age, sex, smoking habits, body mass index, socio-professional category and cleaning products. Model 5: after excluding participants who lived less than one year at the same address (N=46) and adjusted for age, sex, smoking habits, body mass index, socio-professional category and cleaning products.

[#]Estimates were also adjusted for background NO₂.

[†]Results in bold represent significant results (p-values≤0.05).

Table 3. Associations between outdoor air pollution and exhaled 8-iso concentration among participants without asthma.

		NO ₂	NO _x	PM ₁₀	PM _{2.5}	Total traffic load on major roads in a 100-m buffer of the home [#]	Traffic intensity at the road nearest to a participant's home [#]	O ₃	O ₃ -summer
Model 1	n	253	253	185	185	253	253	250	250
	β crude (95% CI)	0.05 (-0.008,0.11)	0.03 (-0.02,0.08)	0.13 (-0.10,0.36)	0.27 (0.05,0.49)[†]	0.07 (-0.04,0.18)	0.03 (-0.02,0.08)	-0.22 (-0.41,-0.03)[†]	-0.55 (-0.81,-0.28)[†]
	p-value	0.08	0.22	0.30	0.02	0.18	0.14	0.02	0.001
Model 2	n	253	253	185	185	253	253	250	250
	β adjusted (95% CI)	0.04 (-0.02,0.10)	0.02 (-0.03,0.07)	0.12 (-0.11,0.36)	0.25 (0.03,0.46)[†]	0.04 (-0.07,0.15)	0.02 (-0.03,0.07)	-0.20 (-0.38,-0.02)[†]	-0.52 (-0.78,-0.26)[†]
	p-value	0.24	0.59	0.33	0.03	0.50	0.37	0.03	0.002
Model 3	n	253	253	185	185	253	253	250	250
	β adjusted (95% CI)	0.03 (-0.03,0.09)	0.01 (-0.05,0.07)	0.11 (-0.12,0.34)	0.23 (0.005,0.46)[†]	0.03 (-0.09,0.15)	0.02 (-0.02,0.06)	-0.20 (-0.39,-0.01)[†]	-0.52 (-0.77,-0.26)[†]
	p-value	0.29	0.63	0.33	0.04	0.56	0.35	0.04	0.002
Model 4	n	233	233	172	172	233	233	230	230
	β adjusted (95% CI)	0.03 (-0.03,0.10)	0.008 (-0.05,0.07)	0.11 (-0.14,0.36)	0.23 (-0.002,0.46)[†]	0.04 (-0.08,0.16)	0.02 (-0.03,0.07)	-0.23 (-0.43,-0.03)[†]	-0.56 (-0.83,-0.29)[†]
	p-value	0.34	0.79	0.39	0.054	0.53	0.34	0.03	0.001

Data are presented as crude β and 95% confidence intervals (95% CI). The linear regression models were conducted with random effects on familial dependence (level 2) and city (level 3). 8-iso concentration was log10 transformed. Results are expressed per 20 µg/m³ increase of NO_x exposure, per 10 µg/m³ increase of NO₂, PM₁₀, O₃ and O₃-summer exposures, per 5 µg/m³ increase of PM_{2.5} exposure, per 4.000.000 vehicles x m /day increase of total traffic load and per 5.000 vehicles/day increase of traffic intensity.

Model 1: unadjusted; Model 2: adjusted for age, sex and smoking habits; Model 3: adjusted for age, sex, smoking habits and cleaning products. Model 4: after excluding participants who lived less than one year at the same address (N=46) and adjusted for age, sex, smoking habits, body mass index, socio-professional category and cleaning products.

[#]Estimates were also adjusted for background NO₂.

[†]Results in bold represent significant results (p-values≤0.05).

Figure legends

Figure 1. Flow chart of the studied population.

ESCAPE, European Study of Cohorts for Air Pollution Effects; 8-iso, 8-isoprostanes; LD, limit of detection of the 8-iso concentration; EBC, Exhaled Breath Condensate.

Figure 2. Associations between exhaled 8-iso concentration and current asthma.

8iso, 8-isoprostanes; LD, limit of detection of the 8-iso concentration.

The logistic models were conducted with random effects on familial dependence.

Estimates were adjusted for age, sex, smoking habits and body mass index. 8-iso concentration was log10 transformed. Figures are OR (95% confidence intervals) with participants without asthma as reference.

[#]Result is expressed of an increase for one unit of the concentration in 8-iso.

[¶]The 8-iso concentration below the limit of detection was used as reference category.

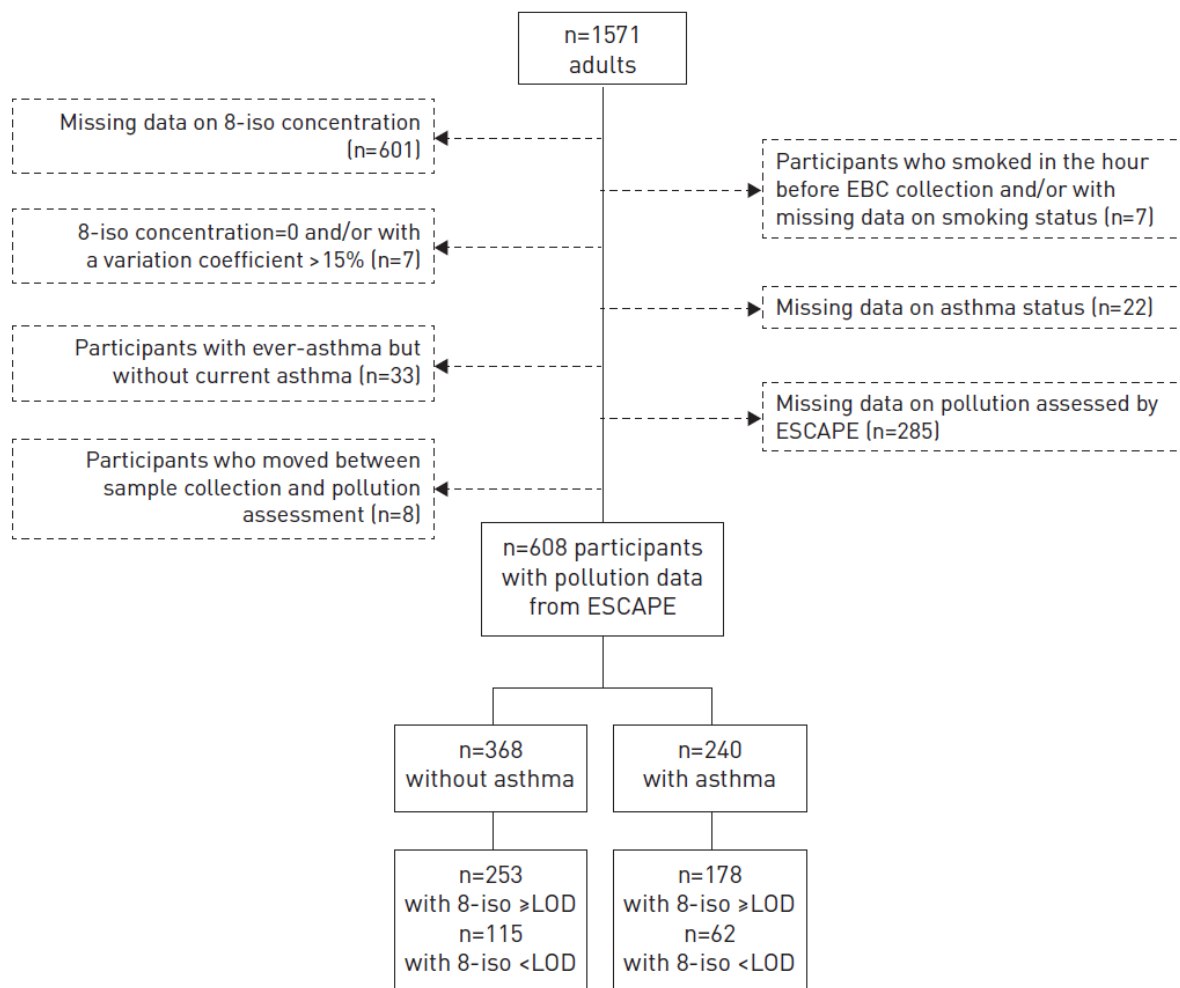


Figure 1

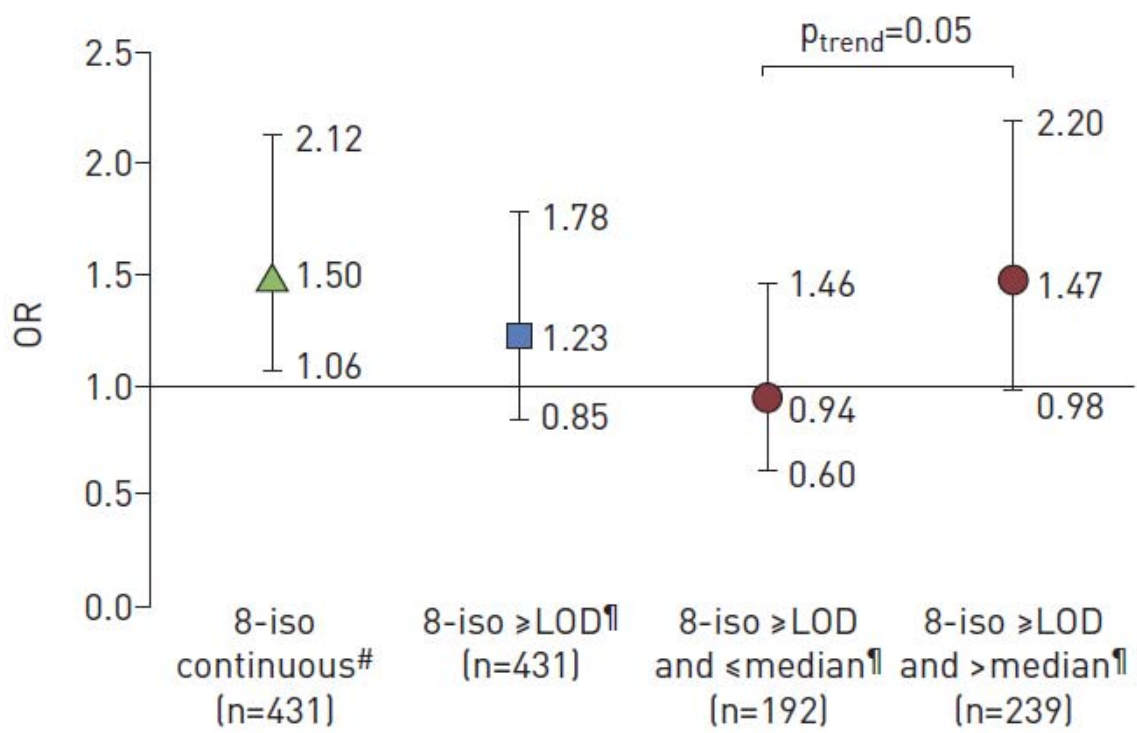


Figure 2